

Targets and calibration

Receptor panel
8 pest targets + *M. sexta* + *B. mori*
(AlphaFold models; Asp189-eq. checked)



Subsite mapping
2PTC / 1SFI → 10 receptors
Foldseek TM-align: 20/20 pairs accepted



Scoring calibration
6 natural inhibitors + 5 shuffled decoys
× 2 receptors (22 systems)
Boltz-2 · ligand RMSD · MM-GBSA

Generation and triage

RFdiffusion, cyclic
11 lengths (5-20 aa) × 10 × 8 targets
880 backbones



ProteinMPNN
22,066 unique sequences



Cleavage-motif screen
(circular, 7 rules)
1,829 resistant-like (8.3%)



Boltz-2 co-folding
(cyclic peptide + receptor MSA)
confidence = 0.8 pLDDT + 0.2 ipTM



Top-1 per target
(highest confidence among
resistant-like)



MD 50 ns, 1 replicate
(linear topology, pH 10)
S1 occupancy · catalytic contacts
· peptide RMSD

Not evaluated here: selectivity against non-target proteases; enzymatic activity; macrocycle closure in the MD topology.